SOLVED PRACTICE WORKBOOK

Geography 03 - Vulcanism and Earthquakes / India Seismic Zones - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Geography 03 - Vulcanism and Earthquakes / India Seismic Zones

BASIC MCQS / REMEDIATION

Practice rule

These are original UPSC-style questions, not claimed PYQs. The renderer applies the stable `geography-03` seed to reposition the correct option while preserving its content. Final keys must be balanced and non-patterned.

MCQ 1 - Endogenic processes

Which statement most accurately describes an endogenic process?

A. It is driven mainly by energy and forces originating within the Earth and includes tectonism, earthquakes and vulcanism. B. It is any process that lowers relief, including only rivers, glaciers and wind. C. It refers exclusively to volcanic eruptions at plate boundaries. D. It is a climatic process caused by unequal solar heating.

Answer: A.

Explanation: Endogenic processes are internally driven. Option B describes exogenic denudation, C is too narrow because earthquakes and non-volcanic tectonism also qualify, and D belongs to atmospheric energy processes.

MCQ 2 - Plate-boundary signature

Which combination is correctly matched?

A. Subduction margin - trench, inclined shallow-to-deep seismic zone and volcanic arc B. Transform margin - deep-focus earthquakes and continuous arc volcanism C. Mid-ocean ridge - predominantly deep earthquakes and rhyolitic calderas D. Continent-continent collision - ocean trench and routine basaltic island arc

Answer: A.

Explanation: A gives the diagnostic subduction system. Transform and ridge earthquakes are mainly shallow. A mature continent-continent collision has consumed the intervening oceanic slab and is not a standard island-arc setting.

MCQ 3 - Normal fault

A normal fault most directly indicates:

A. Crustal compression in which the hanging wall moves up B. Horizontal shear with no vertical component by definition C. Magma-chamber collapse without brittle failure D. Crustal extension in which the hanging wall moves down relative to the footwall

Answer: D.

Explanation: Tensional stress lengthens crust and produces normal faulting. A defines reverse faulting, B describes ideal strike-slip movement, and C describes a caldera process rather than a fault definition.

MCQ 4 - Elastic rebound

Elastic rebound theory explains:

A. Why all aftershocks are smaller than every foreshock B. How slow strain accumulation on a locked fault can be released by sudden rupture and partial recovery C. Why liquid magma necessarily triggers a tectonic earthquake D. How surface waves convert into tides in the ocean

Answer: B.

Explanation: The theory links gradual loading to sudden slip. A is false because sequence labels are retrospective and magnitudes vary. C and D conflate unrelated processes.

MCQ 5 - Earthquake source class

Which event is best classified as reservoir-associated seismicity?

A. Slip promoted near susceptible faults after reservoir loading and pore-pressure change B. A megathrust rupture caused solely by atmospheric pressure C. A cave-roof collapse generating a basin-wide tsunami D. A volcanic ash cloud recorded by a seismometer

Answer: A.

Explanation: Reservoir impoundment can alter loading and pore pressure near pre-existing weaknesses. It does not create plate stress from nothing. The other options confuse source mechanisms and scales.

MCQ 6 - Focal depth

Which statement about focal depth is correct?

A. A shallow earthquake is always more damaging than any deep earthquake. B. Focal depth and epicentral distance are the same quantity. C. Transform faults routinely generate earthquakes at 500-700 km depth. D. Deep-focus earthquakes are characteristic of descending slabs and may occur hundreds of kilometres below the surface.

Answer: D.

Explanation: Intermediate and deep seismicity follows subducting lithosphere. Transform events are shallow. Damage cannot be inferred from depth alone, and depth is vertical source position rather than surface distance.

MCQ 7 - Focus and epicentre

Which distinction is correct?

A. The focus is the worst-damaged town; the epicentre is the largest-magnitude station. B. The focus is the subsurface rupture origin; the epicentre is its vertical projection on the surface. C. The epicentre lies inside the Earth and the focus lies on the surface. D. Both terms refer to the entire seismic zone.

Answer: B.

Explanation: A is the standard geometric distinction. Damage may peak away from the epicentre because of directivity, site effects and construction.

MCQ 8 - Locating an epicentre

Why are observations from at least three seismic stations used in the elementary epicentre-location method?

A. Each station supplies a distance circle, and their intersection estimates the epicentre. B. One station measures longitude, one latitude and one magnitude. C. S waves alone indicate a unique direction at each station. D. Three stations eliminate the need for an Earth-velocity model.

Answer: A.

Explanation: P-S arrival difference gives distance, not a unique bearing. Multiple circles constrain position. Operational location uses more stations and a velocity model.

MCQ 9 - P and S waves

Which statement is correct?

A. S waves arrive before P waves because shear motion is faster. B. P waves move particles only perpendicular to propagation. C. Both P and S waves are confined to the Earth's surface. D. P waves are compressional and can cross liquids, whereas S waves are shear waves requiring a solid medium.

Answer: D.

Explanation: P waves arrive first and travel through solids and liquids. S waves are slower and cannot propagate through liquid. Both are body waves.

MCQ 10 - Surface waves

Which pairing is correct?

A. Love wave - liquid-only; Rayleigh wave - gas-only B. Love wave - mainly horizontal shear; Rayleigh wave - elliptical rolling motion C. Love wave - first arrival; Rayleigh wave - always second arrival D. Love wave - compressional body wave; Rayleigh wave - deep-core shear wave

Answer: B.

Explanation: Love and Rayleigh are surface-wave modes. Love produces strong horizontal motion; Rayleigh produces rolling vertical and horizontal motion.

MCQ 11 - Shadow zones

The absence of direct S waves beyond about 103° from an earthquake primarily indicates that:

A. The outer core is liquid and lacks the shear rigidity needed for S-wave transmission. B. The mantle is entirely molten. C. The inner core is a vacuum. D. S waves travel faster than P waves through liquids.

Answer: A.

Explanation: S-wave extinction is the decisive liquid outer-core evidence. It does not imply a molten mantle or empty inner core.

MCQ 12 - Magnitude and intensity

Which statement is correct?

A. Intensity is constant everywhere for a given event. B. One earthquake has one preferred magnitude estimate but may produce many place-specific intensities. C. Magnitude is assigned from building damage alone. D. Mw and MMI are interchangeable labels for the same quantity.

Answer: B.

Explanation: Magnitude describes source size; intensity describes effects at a location. Mw is a magnitude scale, while MMI is an intensity scale.

MCQ 13 - Moment magnitude

Moment magnitude is physically linked most directly to:

A. Number of media reports and casualties B. Rock rigidity, ruptured fault area and average slip C. Distance from the nearest city only D. Duration of the monsoon

Answer: B.

Explanation: Seismic moment is rigidity × area × slip. Human impact does not define Mw.

MCQ 14 - Global belts

Which global belt is correctly characterised?

A. Circum-Pacific belt - subduction trenches, volcanic arcs, shallow-to-deep earthquakes and tsunami potential B. Mid-Atlantic Ridge - continent-continent collision with no volcanism C. Mediterranean-Himalayan belt - a single uniform oceanic island arc D. Stable shields - completely earthquake-free interiors

Answer: A.

Explanation: A captures the Ring of Fire system. Ridges are divergent and volcanic; the Mediterranean-Himalayan belt is tectonically complex; shields can host intraplate earthquakes.

MCQ 15 - Site amplification

Site amplification most commonly occurs when:

A. A volcano emits only water vapour. B. A fault is absent from the entire region. C. Seismic waves enter softer unconsolidated sediment and selected motions increase relative to nearby rock sites. D. All waves accelerate through soft soil and lose amplitude completely.

Answer: C.

Explanation: Soft sediment and basin geometry can amplify and prolong shaking. The response is frequency-dependent, not a universal simple increase.

MCQ 16 - Liquefaction

Liquefaction is best described as:

A. Dissolution of limestone by rainwater B. Conversion of lava into volcanic glass C. Melting of bedrock due to frictional heat D. Temporary strength loss in loose saturated granular sediment as pore pressure rises during shaking

Answer: D.

Explanation: Liquefaction reduces effective stress; the grains do not literally melt. Floodplains, deltas, reclaimed land and old channels are common susceptible settings.

MCQ 17 - Hazard and disaster

Which formulation is most accurate?

A. Every recorded earthquake is automatically a disaster. B. Vulnerability is the magnitude of the earthquake source. C. A seismic hazard becomes a disaster when it causes severe disruption in an exposed, vulnerable society beyond coping capacity. D. Capacity increases risk by definition.

Answer: C.

Explanation: Hazard, exposure, vulnerability and capacity must be separated. Small uninhabited events may be hazards without disasters.

MCQ 18 - Tsunami source

Which earthquake mechanism is generally most efficient at generating a large basin-wide tsunami?

A. A deep continental strike-slip event with no water displacement B. A minor mine collapse far inland C. A slow atmospheric-pressure change D. A submarine megathrust producing rapid vertical displacement of the sea floor

Answer: D.

Explanation: Rapid vertical displacement of a large sea-floor area moves the whole water column. Strike-slip movement is less efficient unless it triggers a landslide or includes vertical displacement.

MCQ 19 - Tsunami behaviour

As a tsunami enters shallower coastal water, it generally:

A. Converts into an S wave B. Speeds up, lengthens and necessarily disappears C. Slows, shortens in wavelength and grows in height through shoaling D. Becomes a tide controlled by the Moon

Answer: C.

Explanation: Reduced depth lowers speed; energy compression raises wave height. A tsunami is not a tidal wave.

MCQ 20 - Volcano anatomy

Which distinction is correct?

A. A crater is a vent-centred depression; a caldera is a much larger collapse depression commonly linked to substantial magma withdrawal. B. A caldera is always a small side vent. C. A magma chamber is located only above the crater. D. A conduit is an erosional river channel.

Answer: A.

Explanation: Caldera collapse is structurally and dimensionally distinct from an ordinary crater. Chambers and conduits are subsurface parts of the magma system.

MCQ 21 - Silica and viscosity

All else equal, increasing magma silica commonly:

A. Guarantees a harmless effusive eruption B. Eliminates dissolved volatiles C. Converts magma into sedimentary rock before ascent D. Increases polymerisation and viscosity, making gas escape more difficult

Answer: D.

Explanation: Silica-rich structures impede flow and degassing. Composition controls tendency, not certainty.

MCQ 22 - Explosive eruption

Which chain most directly favours explosive fragmentation?

A. No volatiles -> increasing gas pressure B. Viscous magma -> inhibited degassing -> gas expansion during ascent -> fragmentation C. Falling pressure -> gases remain permanently dissolved D. Fluid basalt -> open degassing -> guaranteed Plinian column

Answer: B.

Explanation: Exsolution and trapped gas create overpressure. Fluid magma with open pathways usually degasses more easily.

MCQ 23 - Dyke and sill

Which pairing is correct?

A. Dyke - surface ash deposit; sill - tsunami wave B. Dyke - concordant dome; sill - deep batholith only C. Dyke - cuts across layers; sill - intrudes approximately parallel to layers D. Dyke - crater; sill - caldera

Answer: C.

Explanation: Relation to country-rock layering is the close-option distinction.

MCQ 24 - Volcanic products

Which one is correctly matched?

A. Pyroclastic density current - cool groundwater moving slowly uphill B. Lahar - water-mobilised volcanic debris flow, commonly channelled down valleys C. Lapilli - volcanic gas only D. Volcanic bomb - ash particle smaller than 2 mm

Answer: B.

Explanation: Lahars are sediment-water flows. Density currents are hot gas-particle flows. Lapilli are 2-64 mm fragments; bombs/blocks are coarse.

MCQ 25 - Eruption style

Which eruption style is most closely associated with sustained high eruption columns and widespread tephra?

A. Icelandic fissure B. Quiet effusive degassing only C. Plinian D. Hawaiian

Answer: C.

Explanation: Plinian activity is highly explosive. Hawaiian and Icelandic styles are generally more effusive.

MCQ 26 - Volcano morphology

Which description best fits a shield volcano?

A. A steep cone made only of glacial ice B. Broad gentle slopes built mainly by repeated low-viscosity basaltic flows C. A fold mountain produced without magma D. A deep ocean trench

Answer: B.

Explanation: Fluid lava travels far and builds a low-angle profile. Composite volcanoes are generally steeper and layered.

MCQ 27 - Activity status

Which statement is correct?

A. Active, dormant and extinct are evidence-based status descriptions, not a compulsory one-way life cycle. B. A shield volcano can never be active. C. Dormant means eruption is physically impossible. D. Every volcano inactive for one century is automatically extinct.

Answer: A.

Explanation: Geological, geophysical and gas evidence matters. Long repose may still end in renewed activity.

MCQ 28 - Ring of Fire

The Circum-Pacific “Ring of Fire” is best understood as:

A. A perfect circle of only basaltic shield volcanoes B. A climatic belt created by ocean currents C. A continent-interior rift with no earthquakes D. A linked system of trenches, subducting slabs, earthquake zones and volcanic arcs around much of the Pacific margin

Answer: D.

Explanation: The Ring of Fire is a geophysical system, not a mathematically perfect ring or a single volcano type.

MCQ 29 - Divergent volcanism

Magma at a mid-ocean ridge is generated mainly because:

A. Continents collide and completely stop mantle flow B. Surface ash is buried by glaciers C. A cold slab adds water to a mantle wedge D. Hot mantle rises and partially melts as pressure decreases

Answer: D.

Explanation: Decompression melting is the principal ridge mechanism. Slab-derived flux melting describes subduction arcs.

MCQ 30 - Subduction volcanism

At a subduction zone, magma generation is promoted when:

A. The oceanic plate becomes less dense than the atmosphere B. Volatiles released from the descending slab lower the melting temperature in the overlying mantle wedge C. The trench removes all water from minerals D. The transform fault creates a mid-ocean ridge

Answer: B.

Explanation: Flux melting above the slab feeds volcanic arcs.

MCQ 31 - Mantle plume

Which is the characteristic plume/hot-spot prediction?

A. A plume tail always creates continent-continent collision. B. Hot-spot chains become younger away from the active end in both directions. C. Every hot spot lies exactly on a trench. D. A moving plate over a relatively persistent source can create an age-progressive volcanic chain.

Answer: D.

Explanation: Age generally increases away from the active hot-spot end. The hypothesis is useful but scientifically qualified.

MCQ 32 - Flood basalt

Which statement about the Deccan Traps is correct?

A. They consist only of limestone caves. B. They are a continental flood-basalt province built mainly by extensive fissure eruptions, not a single giant cone. C. They were formed by a modern tsunami. D. They are an active Himalayan island arc.

Answer: B.

Explanation: Repeated basalt sheets produce plateau and step-like relief.

MCQ 33 - Volcanic climate forcing

Temporary global or hemispheric cooling after a major explosive eruption is most directly associated with:

A. Stratospheric sulphate aerosols reflecting incoming sunlight B. Coarse bombs remaining suspended for decades C. Lava absorbing all ocean heat D. S waves entering the atmosphere

Answer: A.

Explanation: Sulphur gases converted to fine aerosols in the stratosphere provide the main short-term radiative pathway.

MCQ 34 - Constructive effect

Which is a constructive long-term effect of vulcanism?

A. Guaranteed absence of future hazards B. Elimination of all atmospheric gases C. Conversion of every volcanic slope into a desert D. Creation of new crust and land, geothermal potential, hydrothermal minerals and weathered fertile material

Answer: D.

Explanation: Benefits coexist with episodic risk; they do not remove it.

MCQ 35 - India tectonic setting

Which regional pairing is correct?

A. Kachchh - mid-ocean ridge spreading axis B. Himalaya - routine oceanic island arc after complete subduction C. Andaman-Nicobar - Sunda subduction/island-arc setting with earthquake, tsunami and volcanic links D. Peninsular India - absolutely aseismic shield

Answer: C.

Explanation: Andaman is the subduction/island-arc system. Kachchh is an intraplate rift-reactivation setting, Himalaya a continental collision, and the peninsula is relatively stable but not aseismic.

MCQ 36 - Bhuj lesson

The 2001 Bhuj earthquake is most useful for demonstrating:

A. That all major Indian earthquakes occur only in the Himalaya B. That liquefaction occurs only in solid granite C. Severe intraplate risk in Kachchh combined with vulnerable buildings and local ground effects D. That seismic zones predict event dates

Answer: C.

Explanation: Bhuj corrects the boundary-only and stable-shield assumptions and illustrates the risk equation.

MCQ 37 - Peninsular exceptions

Which pair corrects the statement "Peninsular India is earthquake-free"?

A. Kīlauea and Mauna Loa B. Andes and Aleutians C. Barren Island and Narcondam D. Koyna and Latur

Answer: D.

Explanation: Koyna and Latur are standard Indian intraplate/reservoir-associated examples.

MCQ 38 - Seismic zonation

Which statement about seismic zonation is correct?

A. It guarantees equal shaking everywhere inside one zone. B. It replaces structural design and construction quality. C. It is a broad hazard/design classification and does not predict the time, place and magnitude of the next earthquake. D. It is identical to an intensity map of one past event.

Answer: C.

Explanation: Macrozonation supports design but cannot resolve local site response or predict events.

MCQ 39 - Microzonation

Seismic microzonation primarily adds:

A. Finer site-specific information on geology, soil, ground motion, liquefaction and related local effects B. A list of volcanic eruption styles C. A guaranteed earthquake date for each ward D. A replacement for all building codes

Answer: A.

Explanation: It is a local hazard/risk-planning tool, not prediction.

MCQ 40 - Early warning

Earthquake early warning differs from prediction because it:

A. names the exact future event years before rupture. B. prevents the earthquake from occurring. C. Detects a rupture that has already begun and may alert more distant places before strong shaking arrives. D. works equally well at the epicentre with unlimited warning time.

Answer: C.

Explanation: Warning time is short and near-source blind zones remain. EEW can automate protective action but cannot stop the source.

MCQ 41 - Remedial: magnitude versus intensity

A learner writes, "The earthquake had intensity Mw 7.0 everywhere." What is the best correction?

A. Intensity and magnitude are identical, but the number must be rounded. B. Mw is a magnitude; intensity is place-specific and should be expressed through an intensity scale or observed effects. C. Mw measures only casualties. D. Every place has the same intensity for one event.

Answer: B.

Explanation: The sentence mixes event size and local effect. One event may produce a mapped gradient of intensities.

MCQ 42 - Remedial: S waves

Why do S waves not pass through the liquid outer core?

A. Liquids always absorb P waves but transmit S waves. B. S waves are electromagnetic. C. S waves are tides. D. A liquid cannot sustain the shear stress required for S-wave propagation.

Answer: D.

Explanation: Shear rigidity is the controlling property.

MCQ 43 - Remedial: epicentre

Which statement avoids the epicentre trap?

A. The epicentre is always the place with the most deaths. B. The epicentre is a geometric surface projection; the greatest loss may occur elsewhere because site and vulnerability modify shaking. C. The epicentre is the entire fault plane. D. The epicentre measures magnitude.

Answer: B.

Explanation: Damage geography and source geometry must be separated.

MCQ 44 - Remedial: hazard and disaster

Which situation is a seismic hazard without necessarily being a disaster?

A. Any earthquake listed by a seismic network B. A building-code document C. A strong remote earthquake that causes no severe social disruption because exposure is minimal D. A dormant fault with no possibility of movement

Answer: C.

Explanation: Disaster requires realised severe impact on society.

MCQ 45 - Remedial: tsunami terminology

Why is “tidal wave” an unsafe synonym for tsunami?

A. Tsunamis are generated by sudden water displacement and are not caused by astronomical tides. B. Tides are seismic body waves. C. Tsunamis cannot cross an ocean. D. Tsunamis occur only at low tide.

Answer: A.

Explanation: The source mechanism is unrelated to lunar/solar tides.

MCQ 46 - Remedial: shield versus composite

Which comparison is correct?

A. Shield volcanoes are always extinct and composite volcanoes always active. B. Both terms classify focal depth. C. Shield volcanoes are broad products of fluid flows; composite volcanoes are layered cones commonly associated with mixed and explosive activity. D. Composite volcanoes form only at mid-ocean ridges.

Answer: C.

Explanation: Form and status are separate classifications.

MCQ 47 - Remedial: zone map

A city lies in one national seismic zone. Which conclusion is justified?

A. Every neighbourhood has identical soil and shaking. B. The next earthquake date is known. C. The zone supplies a broad design-hazard input, but local microzonation and code compliance are still required. D. Retrofitting is unnecessary.

Answer: C.

Explanation: Macrozones do not erase local geology or building differences.

MCQ 48 - Remedial: prediction versus preparedness

Which policy follows from the present inability to predict earthquake time deterministically?

A. Prioritise enforced resilient construction, retrofit, lifeline continuity, monitoring, drills and public education. B. Build only after an earthquake warning is issued. C. Treat low recent seismicity as proof of permanent safety. D. Stop seismic monitoring.

Answer: A.

Explanation: Risk reduction manages consequences under uncertainty; absence of recent events does not erase accumulated hazard.

PYQS AND ANSWER PRACTICE

PYQ control note

- Exact wording is reproduced from the routed local UPSC question papers.
- UPSC does not publish model Mains answers; the Mains solutions below are independent examiner-grade models.
- 2018 and 2023 Prelims keys are unavailable in the local verified key set; answers are explicitly labelled inferred.
- The 2024 Set-A answer is verified from the local official key.
- The 2026 Set-A key is local and provisional. It is reported as provisional, never as the final UPSC key and never replaced by an inference.

VERIFIED PYQ 1 - UPSC Prelims 2018 GS-I Q85: Barren Island

Exact question

Consider the following statements:

1. The Barren Island volcano is an active volcano located in the Indian territory.
2. Barren Island lies about 140 km east of Great Nicobar.
3. The last time the Barren Island volcano erupted was in 1991 and it has remained inactive since then.

Which of the statements given above is/are correct?

A. 1 only B. 2 and 3 only C. 3 only D. 1 and 3 only

INFERRED ANSWER - NOT OFFICIALLY VERIFIED LOCALLY: A, 1 only. Confidence: high.

Explanation

- Statement 1 is correct: Barren Island in Indian territory is the country's active-volcano example.
- Statement 2 is false: the location relative to Great Nicobar is wrong; Barren Island lies in the northern Andaman Sea, northeast of Port Blair, not about 140 km east of Great Nicobar.
- Statement 3 is false: activity resumed after 1991; therefore “remained inactive since then” is incorrect.

Why this PYQ matters: UPSC combines activity status, map location and eruption chronology. “Active” is not equivalent to “erupting continuously.”

VERIFIED PYQ 2 - UPSC Prelims 2023 GS-I Q65: P and S waves

Exact question

Consider the following statements:

1. In a seismograph, P waves are recorded earlier than S waves.
2. In P waves, the individual particles vibrate to and fro in the direction of wave propagation, whereas in S waves, the particles vibrate up and down at right angles to the direction of wave propagation.

Which of the statements given above is/are correct?

A. 1 only B. 2 only C. Both 1 and 2 D. Neither 1 nor 2

INFERRED ANSWER - NOT OFFICIALLY VERIFIED LOCALLY: C, both 1 and 2. Confidence: high.

Explanation

P waves are faster compressional body waves, so they arrive first. Their particle motion is parallel to propagation. S waves are shear waves with particle motion perpendicular to propagation. The statement's “up and down” language illustrates the transverse relation; S polarisation may occur in different perpendicular orientations.

VERIFIED PYQ 3 - UPSC Prelims 2024 GS-I Q3: products of eruptions

Exact question

Consider the following:

1. Pyroclastic debris
2. Ash and dust
3. Nitrogen compounds
4. Sulphur compounds

How many of the above are products of volcanic eruptions?

A. Only one B. Only two C. Only three D. All four

OFFICIALLY VERIFIED LOCAL SET-A KEY: D, all four.

Explanation

Explosive eruptions produce pyroclastic debris, ash and dust. Volcanic emissions include sulphur compounds and can include nitrogen species/compounds. The official key controls the objective answer. The elimination lesson is that eruption products include gases and chemical species, not only visible lava and rock fragments.

VERIFIED PYQ 4 - UPSC Prelims 2026 GS-I Q31: Tungurahua

Exact question

Tungurahua Volcano, which was declared a Global Geopark by UNESCO in 2025, is situated in which one among the following countries?

A. Ecuador B. Peru C. Bolivia D. Colombia

LOCAL PROVISIONAL SET-A KEY: A, Ecuador. Final official UPSC key not verified.

Status explanation: UNESCO's official 2025 material identifies Tungurahua Volcano UNESCO Global Geopark in Ecuador. The local answer-key document is expressly provisional; this package records its status and does not infer or upgrade it to a final key.

Map trap: Ecuador and Colombia are both Andean states on the Ring of Fire. Tungurahua belongs to Ecuador's volcanic arc.

VERIFIED PYQ 5 - UPSC Mains GS-I 2018 Q6, 10 marks

Exact question: “Define mantle plume and explain its role in plate tectonics.” *(Answer in 150 words.)*

Demand decoding

“Define” requires plume anatomy; “explain its role” requires hot-spot chains, flood basalts and the plate-motion reference frame. A generic answer on plate boundaries misses the demand.

Model answer

A **mantle plume** is a hypothesised buoyant upwelling of anomalously hot mantle material that rises substantially independently of ordinary plate-boundary circulation.

```
deep thermal anomaly -> broad plume head -> flood basalt
                     -> persistent tail -> hot spot
moving plate over tail -> age-progressive volcanic chain
```

Its role in plate tectonics is threefold.

1. **Intraplate volcanism:** It explains volcanoes away from plate margins, such as the Hawaiian hot-spot chain.

2. **Large igneous provinces:** Arrival of a broad plume head may generate voluminous fissure eruptions; the Deccan Traps are the Indian illustration.
3. **Plate-motion record:** Progressive ageing away from an active hot spot indicates the direction and approximate relative rate of plate movement, independently checking evidence from magnetic stripes.
4. **Rifting link:** Thermal doming and magmatism may assist continental break-up, although plate stresses also matter.

The hypothesis is not a complete replacement for plate tectonics. Plumes add a vertical mantle component to a system dominated by horizontal plate interaction, and some intraplate volcanism may have shallower causes.

Why this earns marks

It gives a direct definition, one process diagram, two named examples, four roles and a scientific qualification instead of describing ordinary arc volcanoes.

VERIFIED PYQ 6 - UPSC Mains GS-I 2020 Q4, 10 marks

Exact question: “Discuss the geophysical characteristics of Circum-Pacific Zone.” *(Answer in 150 words.)*

Demand decoding

“Geophysical characteristics” requires tectonic structure, earthquake-depth pattern, volcanism, landforms and tsunami potential-not a list of Pacific countries.

Model answer

The Circum-Pacific Zone or **Ring of Fire** is the world's dominant linked belt of subduction-related seismicity and volcanism around much of the Pacific margin.

```
oceanic plate -> trench -> inclined Wadati-Benioff seismic zone
                          -> mantle-wedge melting -> volcanic arc
```

Its principal characteristics are:

- **Deep trenches** such as Peru-Chile, Japan and Mariana mark descending plates.
- **Shallow, intermediate and deep earthquakes** outline subducting slabs; powerful megathrust events occur on locked plate interfaces.
- **Volcanic arcs** form as slab-derived volatiles promote mantle-wedge melting. Continental arcs occur in the Andes/Cascades and island arcs in Japan, the Philippines and Aleutians.
- **Explosive stratovolcanism** is common because arc magmas are often volatile-rich and relatively viscous.
- **Tsunami potential** is high where submarine megathrust rupture displaces the sea floor.
- Transform segments and back-arc basins add local complexity.

Thus, the Ring is not merely a concentration of volcanoes but a trench-slab-earthquake-arc system produced by continuing Pacific-margin convergence.

Why this earns marks

The answer reconstructs the entire geophysical system, uses map-ready examples and ends with a mechanism-led conclusion.

VERIFIED PYQ 7 - UPSC Mains GS-I 2021 Q7, 10 marks

Exact question: “Mention the global occurrence of volcanic eruptions in 2021 and their impact on regional environment.” *(Answer in 150 words.)*

Demand decoding

The word “mention” requires multiple named occurrences, but “impact on regional environment” requires a short mechanism and consequence for each rather than a bare list.

Model answer

Volcanic eruptions in 2021 occurred across several tectonic settings, especially subduction arcs and hot spots.

Occurrence	Setting/product	Regional environmental impact
La Soufrière, St Vincent	Explosive island-arc eruption	Ash affected air, water, crops and inter-island activity; pyroclastic and lahar hazards threatened valleys
Nyiragongo, DR Congo	Fluid lava from the East African Rift region	Lava affected settlements and infrastructure around Goma; evacuation and seismic unrest disrupted livelihoods
Cumbre Vieja, La Palma, Spain	Intraplate Canary hot-spot volcanism	Lava buried settlements and plantations; ash disrupted air quality and aviation
Semeru, Indonesia	Subduction-arc stratovolcano	Pyroclastic flows, ash and rain-remobilised debris affected valleys and communities

The impacts were therefore spatially selective: wind controlled ash, relief channelled density currents and lahars, and island/urban exposure magnified disruption. Yet, over longer periods, volcanic materials may create new land and fertile weathered soils. Regional impact depends as much on settlement, drainage and wind as on eruption magnitude.

Why this earns marks

It names globally distributed examples, ties each to a product/pathway and provides the constructive counterpoint without diluting the hazard analysis.

SUPPLEMENTARY CROSS-OWNED PYQ 8 - UPSC Mains GS-I 2025 Q7, 10 marks

Exact question: “What are Tsunamis? How and where are they formed? What are their consequences? Explain with examples.” *(Answer in 150 words.)*

Ownership note: The institutional disaster-management route is owned elsewhere, but the physical geography is directly relevant and therefore solved here.

Model answer

A **tsunami** is a train of long-period gravity waves generated by sudden displacement of a large water column; it is not a tide or ordinary wind wave.

Formation and location: Submarine megathrust earthquakes at subduction zones are the chief cause. Rapid vertical sea-floor movement displaces water, producing low-amplitude, fast waves in the deep ocean. As they enter shallow water, speed and wavelength fall while height increases through shoaling. Volcanic collapse, submarine landslides and rare impacts can also generate tsunamis. Major source belts include the Pacific Ring of Fire and the Sunda-Andaman subduction zone.

Consequences: Run-up and inundation cause drowning, erosion, debris impact and infrastructure loss; saline water contaminates soils and groundwater; ports, fisheries, tourism and coastal livelihoods suffer. Funnel-shaped bays, low slopes and dense unplanned settlements magnify loss.

The 2004 Sumatra-Andaman earthquake devastated Indian Ocean coasts; the 2011 Tōhoku event demonstrated earthquake-tsunami-nuclear cascading risk. Warning systems reduce exposure, but near-source coasts must also use natural warning signs and rapid evacuation.

Why this earns marks

Every sub-demand-definition, how, where, consequences and examples-is visibly answered.

ORIGINAL 10-MARK QUESTION 1

Question: Explain why a moderate earthquake may cause greater damage than a larger earthquake. *(150 words.)*

Model answer

Magnitude measures source size, not automatic damage. A moderate event may be more destructive when the complete risk chain is unfavourable:

1. **Shallow source and proximity:** less energy is lost before waves reach a settlement.
2. **Directivity:** rupture may focus strong motion towards the city.
3. **Soft ground:** alluvium and basin sediments amplify selected frequencies and prolong shaking.
4. **Liquefaction:** loose saturated sand loses strength, damaging foundations and utilities.
5. **Resonance:** dominant ground period may match vulnerable building heights.
6. **Weak building stock:** unreinforced masonry, soft storeys, poor connections and heavy unanchored elements fail.
7. **Exposure and cascading failure:** dense population, fire, blocked roads and lifeline disruption multiply loss.

```
damage ≠ magnitude alone
damage = source × path × site × exposure × vulnerability ÷ capacity
```

Thus a larger deep earthquake beneath rock and code-compliant construction may cause less loss than a smaller shallow event beneath saturated alluvium and non-engineered buildings. Earthquake policy must therefore combine source hazard with microzonation, resilient construction and retrofit.

ORIGINAL 10-MARK QUESTION 2

Question: Distinguish between volcanic ash, pyroclastic density currents and lahars, and explain why they require different hazard maps. *(150 words.)*

Model answer

All three contain volcanic material but move through different media and pathways.

Hazard	Composition/motion	Control	Hazard-map implication
Ash fall	Fine airborne particles settling from plume	Eruption-column height and wind	Broad downwind sectors; aviation, crops, water and roof load
Pyroclastic density current	Hot turbulent gas-particle mixture hugging ground	Collapse/blast energy and relief	Proximal flanks and valleys; extreme temperature and speed
Lahar	Water-mobilised ash and debris	Rain, snow/crater water and drainage	River valleys far beyond cone; can recur after eruption

Ash maps must model wind and accumulation thickness. Density-current maps require topographic run-out zones and exclusion areas. Lahar maps must follow drainage channels, bridges, settlements and reservoirs, including post-eruption rainfall scenarios.

Therefore, a single circular “volcano danger zone” is inadequate. Product, transport medium and terrain create distinct footprints; risk communication and evacuation must match each pathway.

ORIGINAL 15-MARK QUESTION 1

Question: Examine the tectonic and geomorphic reasons for the uneven distribution of earthquake risk in India. *(250 words.)*

Model answer

India's earthquake risk is uneven because **hazard sources vary by tectonic region**, while geomorphology and settlement alter the received shaking.

```
tectonic source -> wave path -> local ground -> exposed built environment
```

Tectonic differentiation

- **Himalaya:** continuing Indian-Eurasian convergence loads a broad fold-thrust belt. Shallow thrust earthquakes combine with steep slopes and landslide cascades.

- **North-East/Indo-Burmese region:** the eastern Himalayan syntaxis and Burma/Sunda interaction create complex compression and strike-slip deformation.
- **Andaman-Nicobar:** Sunda subduction creates earthquake, tsunami and volcanic multi-hazard.
- **Kachchh:** inherited rifts and faults can reactivate within the plate; Bhuj 2001 disproves the assumption that only boundaries generate severe risk.
- **Peninsular shield:** lower average deformation, but Koyna and Latur show reservoir-associated and intraplate exceptions.

Geomorphic and exposure differentiation

- Indo-Gangetic and urban-basin alluvium can amplify shaking.
- Loose saturated floodplain, delta and reclaimed deposits are liquefaction-prone.
- Himalayan relief triggers landslides and cuts roads, power and communications.
- Dense cities, vulnerable masonry, soft storeys and weak enforcement convert hazard into disaster.

Qualification: A BIS macrozone is a broad design input, not a prediction or a neighbourhood risk map. Microzonation, building inventory and lifeline analysis are necessary.

India's seismic-risk map is therefore not merely a tectonic map. It is a layered map of active structures, sediment, slope, urban exposure and construction quality.

ORIGINAL 15-MARK QUESTION 2

Question: “Volcanoes are both agents of destruction and foundations of regional development.” Discuss. *(250 words.)*

Model answer

Volcanoes produce acute, spatially differentiated hazards but also long-lived resources and land.

Destructive role

- Lava buries land and infrastructure.
- Pyroclastic density currents cause extreme proximal destruction.
- Ash damages crops, water, roofs, health and aviation over broad downwind areas.
- Lahars repeatedly affect valleys even after eruption.
- Volcanic gases create toxic local conditions; stratospheric sulphate aerosols may temporarily cool climate.
- Eruptions displace populations and disrupt island economies, ports and tourism.

Developmental foundations

- Weathered basalt and ash can form productive soils, as seen in the Deccan and Java, though fresh ash first imposes costs.
- Geothermal heat supports energy and heating in suitable regions.
- Hydrothermal systems concentrate metallic minerals and industrial materials.
- Volcanism constructs islands, plateaux and new crust.
- Distinctive landscapes support tourism and scientific education.

Analytical balance

The benefits are often continuous or recoverable, whereas hazards are episodic but potentially catastrophic. This explains continued settlement: communities exchange recurrent benefits for probabilistic risk. However, development becomes sustainable only when monitoring, exclusion zones, resilient infrastructure, evacuation routes and livelihood diversification prevent the resource advantage from becoming an exposure trap.

Thus volcanism is not “beneficial” or “destructive” in isolation. Regional outcome depends on product pathways, recurrence, governance and the capacity to live with a dynamic landscape.

ORIGINAL 20-MARK QUESTION 1

Question: Critically examine India's seismic-zonation framework as an instrument of urban risk reduction. *(250 words; expanded model supplied.)*

Model answer

Seismic zonation is indispensable because it converts national-scale seismotectonic evidence into a design language, but it is insufficient as a complete urban-risk instrument.

Value

1. **Common regulatory baseline:** BIS zonation and IS 1893 provide broad design-hazard parameters for structural standards.
2. **Prioritisation:** High-hazard regions can prioritise critical infrastructure, schools, hospitals and emergency routes.
3. **Mainstreaming:** A map makes earthquake resistance part of ordinary building regulation rather than post-disaster relief.
4. **Inter-regional consistency:** It enables comparable minimum design expectations across jurisdictions.

Limitations

1. **Scale:** A macrozone cannot resolve soft alluvium, reclaimed land, basin edges, active fault traces or liquefaction within a city.
2. **Hazard-risk gap:** It maps broad hazard, not exposure, building fragility, network dependence or social vulnerability.
3. **Boundary illusion:** Administrative users may treat zone lines as abrupt physical changes.
4. **Prediction misconception:** A zone neither predicts the next event nor guarantees uniform intensity.
5. **Implementation:** Weak local adoption, informal construction, poor supervision and ageing buildings can defeat a sound code.
6. **Current classification caution:** Official public communication in 2025-2026 refers inconsistently to Zone VI and Zone V for the Himalayan belt, while the publicly verifiable BIS design baseline remains the familiar II-V framework. Regulatory answers must cite the edition/date rather than assert an unverified settled change.

Way forward

```
macrozonation -> city microzonation -> parcel/geotechnical check
-> risk-sensitive land use -> code-compliant design
-> inventory + retrofit -> enforcement + audit + drills
```

NCS/MoES hazard work, GSI mapping, BIS codes, NDMA guidance and State/ULB enforcement must operate as one chain. Zonation is therefore the starting map of resilience, not its final guarantee.

ORIGINAL 20-MARK QUESTION 2

Question: Analyse the common tectonic logic and contrasting surface expressions of earthquakes, volcanoes and tsunamis around the Circum-Pacific belt. *(250 words; expanded model supplied.)*

Model answer

The Circum-Pacific belt is unified by plate convergence but differentiated by the way energy and material reach the surface.

Common tectonic logic

Oceanic lithosphere subducts beneath another oceanic plate or a continent. The locked interface accumulates elastic strain; rupture generates earthquakes. The descending slab releases volatiles, promoting mantle-wedge melting and volcanic arcs. Submarine megathrust displacement may move the ocean floor and generate tsunamis.

```
trench
  -> locked megathrust -> earthquake + possible vertical sea-floor displacement
  -> descending slab -> shallow/intermediate/deep seismicity
  -> slab volatiles -> mantle-wedge melt -> volcanic arc
```

Contrasting expressions

- **Earthquakes** are sudden elastic-energy release and occur along the interface, within the slab and in the overriding plate. Their map extends from shallow trench events to deep Wadati-Benioff zones.
- **Volcanoes** are mass and heat transfer. Magma rises landward of the trench, forming island arcs such as Japan/Philippines or continental arcs such as the Andes.
- **Tsunamis** are water-column responses. They require efficient vertical displacement or another rapid mass movement; not every large earthquake produces one.

Regional controls

Arc magma chemistry and gas favour explosive stratovolcanoes. Trench geometry, rupture area and depth influence earthquakes. Bathymetry, bays and coastal slope control tsunami run-up. Transform segments may be highly seismic but weakly volcanic.

Critical qualification

The “Ring” is neither continuous nor mechanically uniform. It is a connected plate-margin system whose hazards share a tectonic origin but require distinct monitoring, maps and preparedness.